

CHAPTER 2: RECTIFIERS WITH THYRISTOR

3. 1 Experiment of Single Phase Half Wave Rectifier

3.1.2 Single Phase Half Wave Rectifier (with Inductive Load)

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For R=100 ohm, L=50 mH



Oscilloscope

Channel 1: Input Voltage

Channel 2: Firing Angel

Channel 3: Output Voltage

Firing angel	Input voltage		Output voltage	
	RMS	MEAN	RMS	MEAN
45°	5.53V	-253mV	3.97V	2.57V
90°	5.55V	-174mV	2.90V	1.42V
135°	5.56	-187mV	1V	482mV

$$V_{AVG} = \frac{V_M}{2\pi} (1 + \cos \alpha) \quad V_{RMS} = \frac{V_M}{2} \left[\frac{1}{\pi} \left(\pi - \alpha + \frac{\sin 2\alpha}{2} \right) \right]^{0.5}$$

$$V_{rms} = 5.53V \quad V_{max} = V_{rms} * \sqrt{2} \quad V_{max} = 7.82V$$

$$\alpha = 45^\circ$$

$$V_{avg} = 2.125V$$

$$V_{rms} = 3.728V$$

$$\alpha = 90^\circ$$

$$V_{avg} = 1.245V$$

$$V_{rms} = 2.765V$$

$$\alpha = 135^\circ$$

$$V_{avg} = 0.365V$$

$$V_{rms} = 1.178V$$



Figure 1 Experiment for 45°



Figure 2 Experiment for 90°



Figure 3 Experiment for 135°

In conclusion, The experiment validates the theoretical understanding of a single-phase half-wave rectifier. The results show good agreement between theory and practice, confirming the rectifier's functionality and the impact of firing angle on output voltages.

Increasing the firing angle:

Decreases V_{avg} and V_{rms} values.

Reduces and narrows the output waveform.

Lowers the total energy delivered to the load.

3. 2 Single Phase Full Wave Rectifier-Bridge Rectifier

3.2.2 Single Phase Half Controlled Rectifier (with Inductive Load)

For R=100 ohm, L=50 mH



Oscilloscope

Channel 1: Input Voltage

Channel 2: Firing Angel

Channel 3: Output Voltage

Firing angel	Input voltage		Output voltage	
	RMS	MEAN	RMS	MEAN
45°	1.75V	-131mV	875mV	386mV
90°	1.75V	-134mV	874mV	416mV
135°	1.75V	-140mV	636mV	392mV

$$V_{AVG} = \frac{V_M}{\pi} (1 + \cos \alpha) \quad V_{RMS} = \frac{V_M}{\sqrt{2}} \left[\frac{1}{\pi} \left(\pi - \alpha + \frac{\sin 2\alpha}{2} \right) \right]^{0.5}$$

$$V_{rms} = 1.75V \quad V_{max} = V_{rms} * \sqrt{2} \quad V_{max} = 2.47V$$

$$\alpha = 45^\circ$$

$$V_{avg} = 1.34V$$

$$V_{rms} = 1.67V$$

$$\alpha = 90^\circ$$

$$V_{avg} = 0.79V$$

$$V_{rms} = 1.245V$$

$$\alpha = 135^\circ$$

$$V_{avg} = 0.23V$$

$$V_{rms} = 0.53V$$

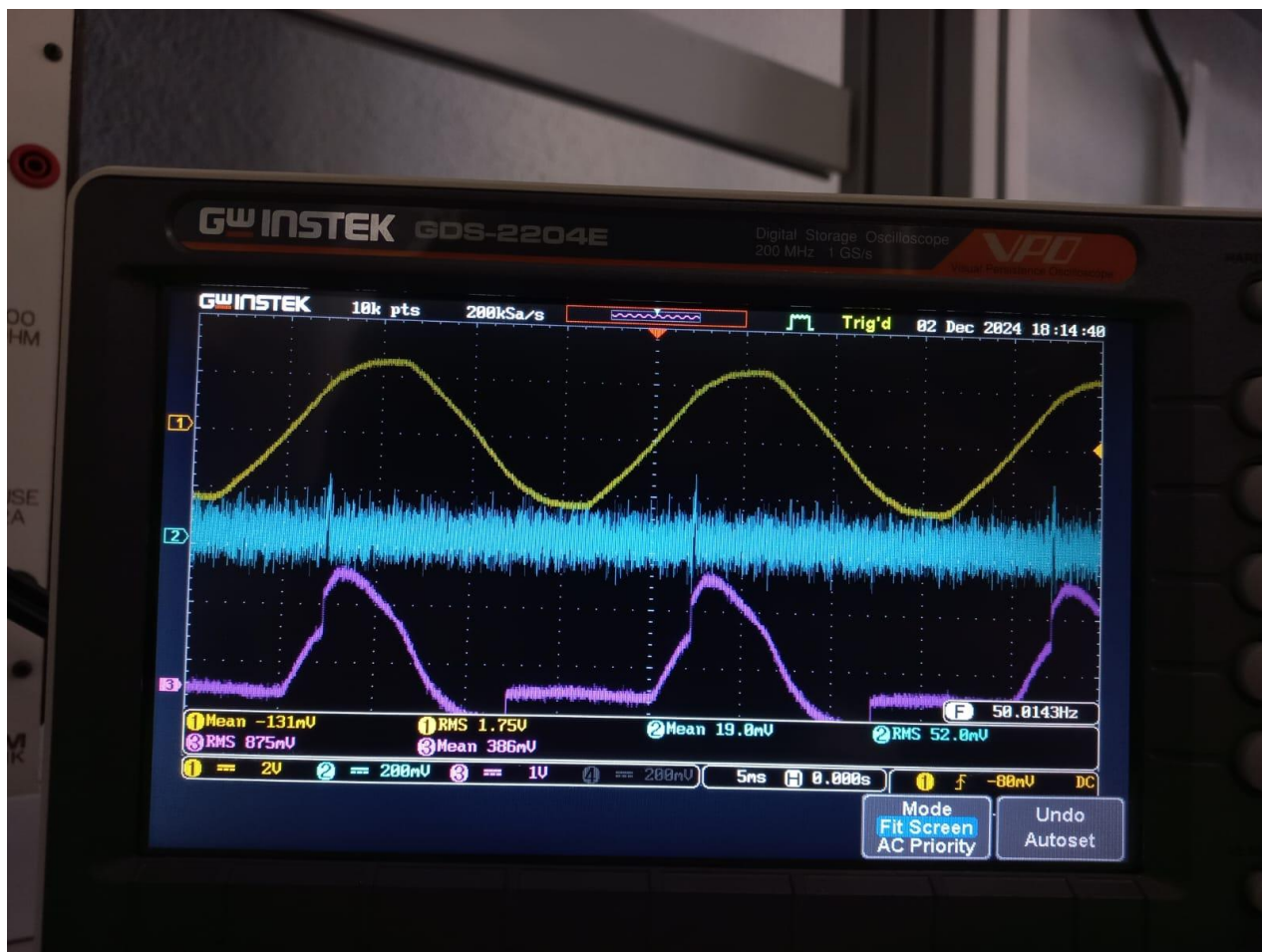


Figure 4 Experiment for 45°

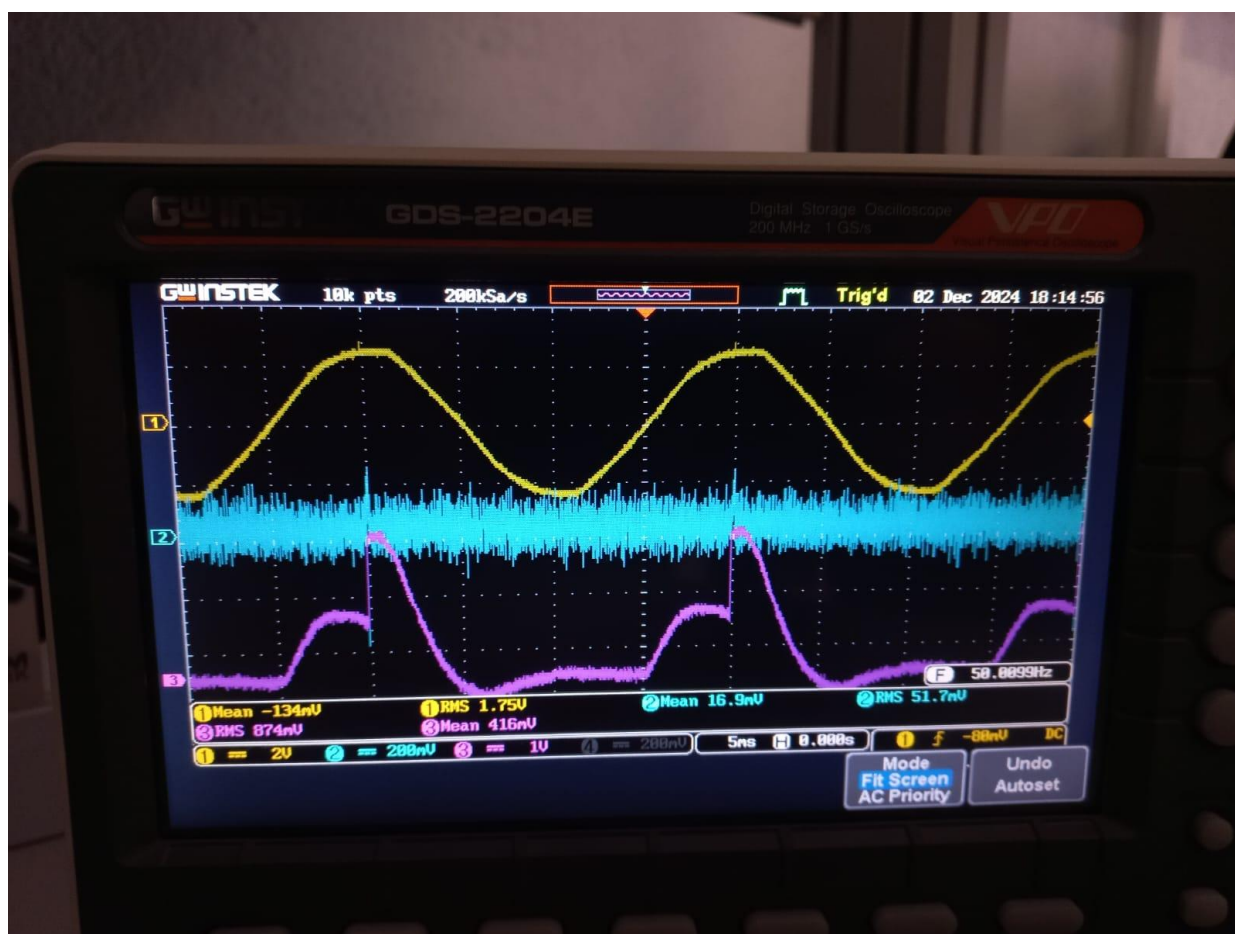


Figure 5 Experiment for 90°



Figure 6 Experiment for 135°

In conclusion, In the Single Phase Half Controlled Rectifier experiment, the rectified output voltages (V_{avg} and V_{rms}) were analyzed for different firing angles (α) using an inductive load. It was observed that as the firing angle α increased, both the average (V_{avg}) and the root mean square (V_{rms}) output voltages decreased. This is because, with a higher α , the thyristor conducts for a shorter duration. As a result, the contribution of the rectified voltage decreases, leading to a lower output voltage.

3.2.4 Single Phase Full controlled rectifier (with Inductive Load)

For R=100 ohm, L=50 mH



Oscilloscope

Channel 1: Input Voltage

Channel 2: Firing Angel

Channel 3: Output Voltage

Firing angel	Input voltage		Output voltage	
	RMS	MEAN	RMS	MEAN
45°	1.88V	-95.8mV	521mV	319mV
90°	1.81V	-183mV	782mV	274mV
135°	1.81V	-99.8mV	513mV	2.84mV

$$V_{AVG} = \frac{2V_M}{\pi} (\cos\alpha) \quad V_{RMS} = \frac{V_M}{\sqrt{2}}$$

Vrms = 1.81V Vmax=Vrms *√2 Vmax =2.56V

α=45°

Vavg= 1.152V

Vrms= 1.81V

α=90°

Vavg=0V

Vrms=1.81V

α=135°

Vavg=-1.152V

Vrms=1.81V



Figure 7 Experiment for 45°

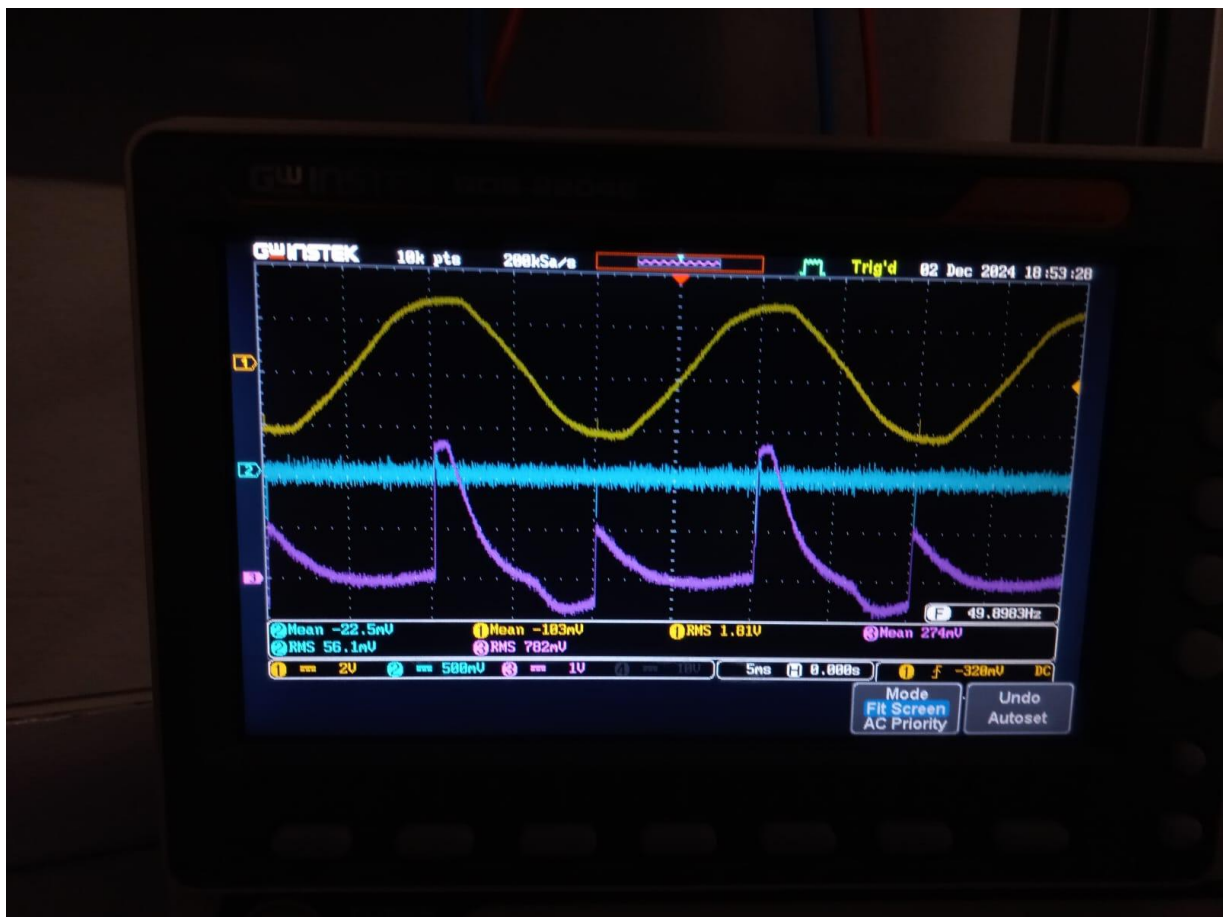


Figure 8 Experiment for 90°

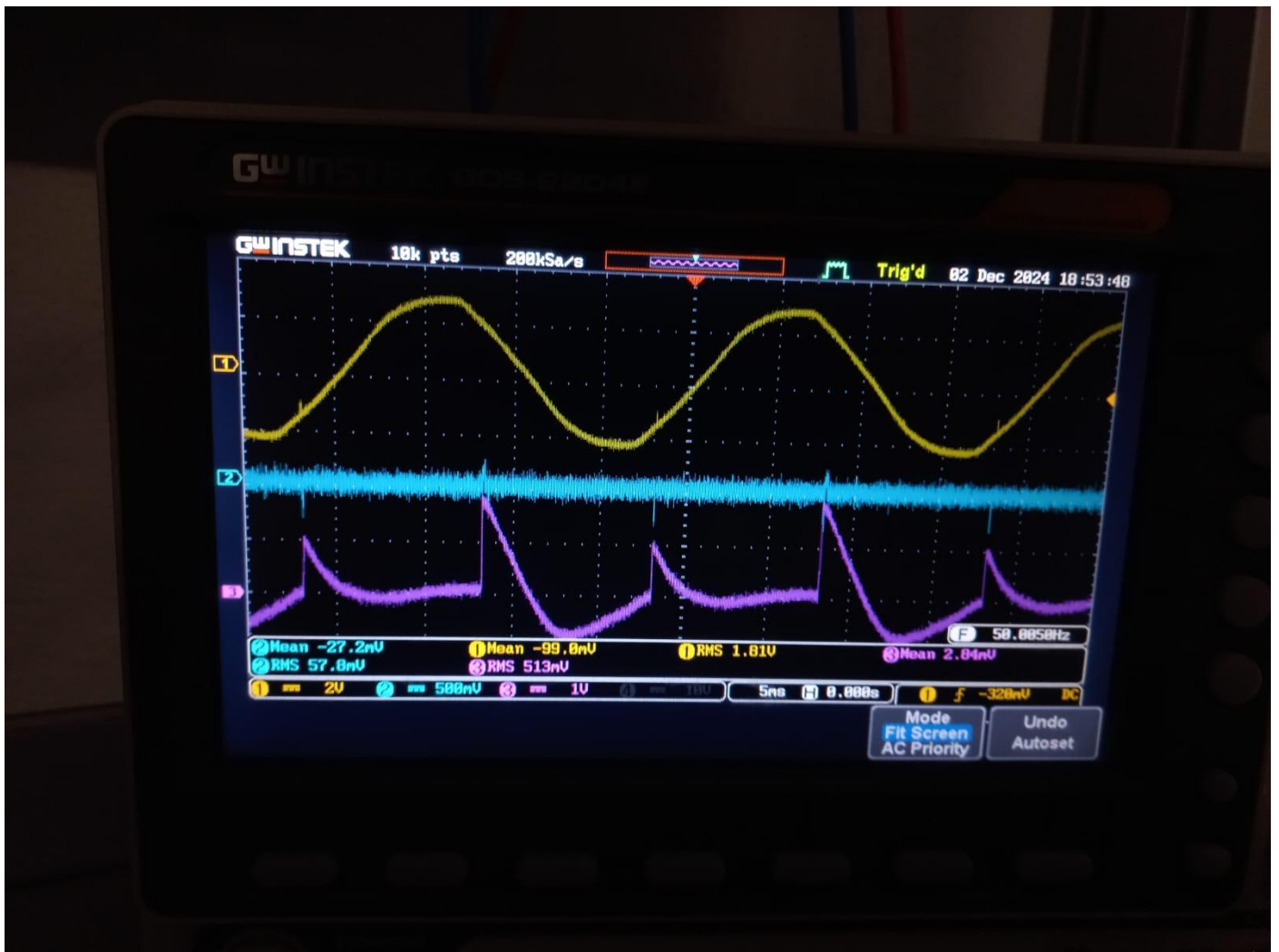


Figure 9 Experiment for 135°

The experimental results generally show the expected trends:

As the angle increases, the V_{avg} value shifts from positive to negative. V_{rms} values should theoretically remain constant, but noticeable differences were observed in the measurements. However, there is no complete overlap between the theoretical and experimental results. This discrepancy is a result of non-ideal effects occurring in real-world conditions, such as the nonlinear behavior of the inductive load and device tolerances.

3. 3 Tri-phase Half-Wave Rectifier

3.3.2 Tri-phase Half Wave Rectifier (with Inductive Load)

For R=200 ohm, L=50 mH



Oscilloscope

Channel 1: Input Voltage

Channel 2: Firing Angel

Channel 3: Output Voltage

Firing angel	Input voltage		Output voltage	
	RMS	MEAN	RMS	MEAN
45°	5.55V	-69mV	5.62V	4.72V
90°	5.55V	-54.6mV	4.17V	2.77V
135°	5.58V	-39.5mV	747mV	418mV

$$V_{AVG} = \frac{3\sqrt{3}V_M}{2\pi} \cos\alpha \quad V_{RMS} = \sqrt{3}V_M \left(\frac{1}{6} + \frac{\sqrt{3}}{8\pi} \cos 2\alpha \right)^{0.5}$$

$$V_{rms} = 5.55V \quad V_{max} = V_{rms} * \sqrt{2} \quad V_{max} = 7.849V$$

$$\alpha = 45^\circ$$

$$V_{avg} = 4.59V$$

$$V_{rms} = 5.55V$$

$$\alpha = 90^\circ$$

$$V_{avg} = 0V$$

$$V_{rms} = 4.25V$$

$$\alpha = 135^\circ$$

$$V_{avg} = -4.59V$$

$$V_{rms} = 5.55V$$



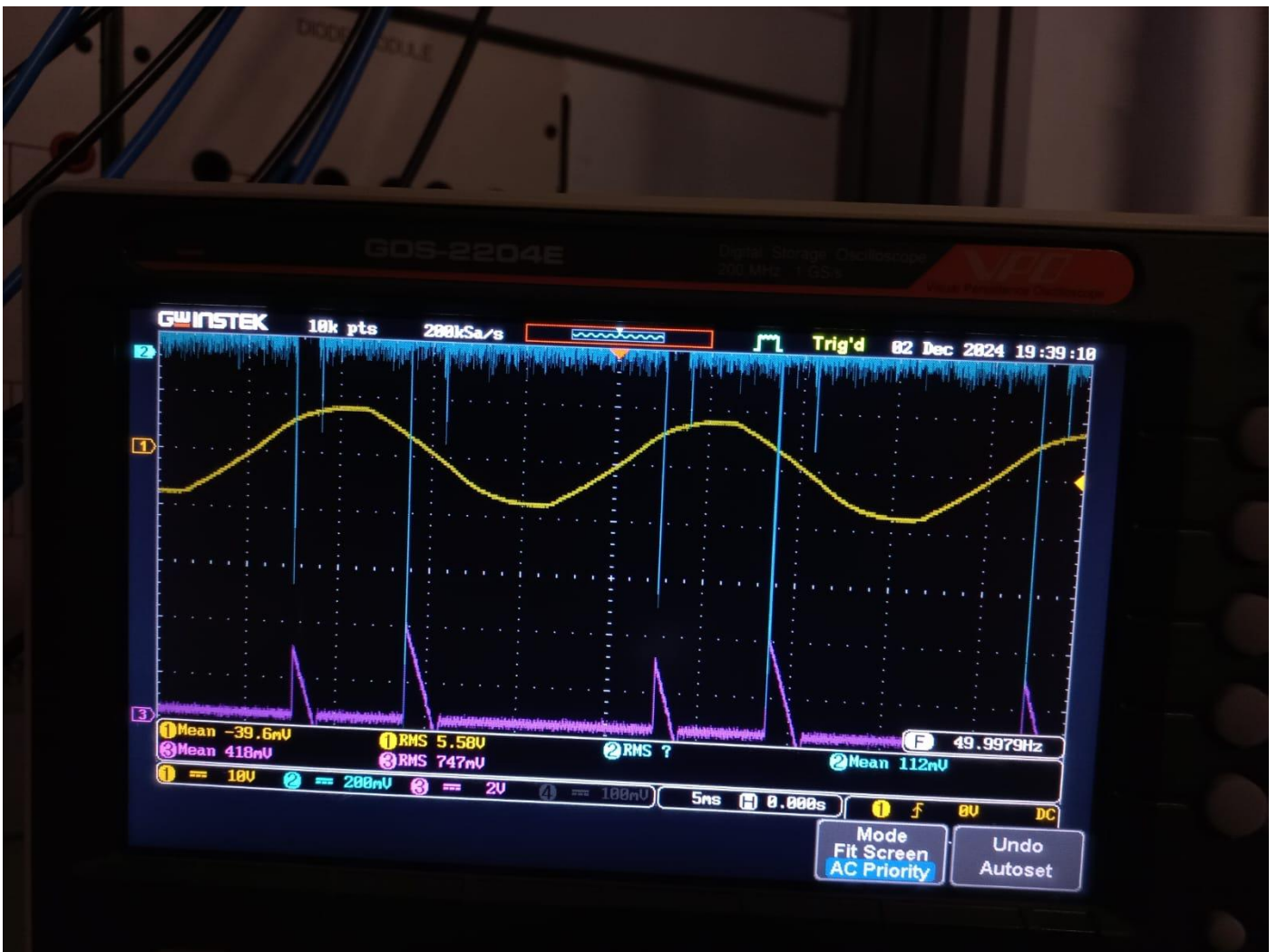


Figure 12 Experiment for 135°

In conclusion ,Effect of Increasing Firing Angle (α): As α increases, the conduction period decreases, leading to a reduction in V_{avg} and V_{rms} . This behavior aligns well with theoretical predictions.

Theoretical vs. Experimental Alignment: While the results for $\alpha=45^\circ$ closely match theoretical values, significant deviations are observed at higher firing angles, especially for $\alpha=135^\circ$. These discrepancies may arise due to:

Non-ideal circuit conditions, such as diode forward voltage drop or imperfect switching behavior.

Measurement errors or limitations in the accuracy of the equipment used.

The influence of the inductive load, which may not fully smooth the waveform due to insufficient inductance or high ripple.